

Week 17: Compress the Data, Launch the Research

Principal Component Analysis and the First Research Proposal

Instructor / Mentor

Kiki Research Training Program

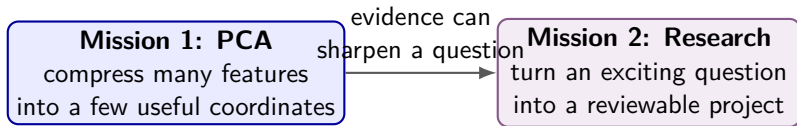
Week 17 of the Revised 22-Week Program

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Two Missions for One Week

Today's Two Missions



The shared habit

Reduce complexity without hiding what matters: PCA does this for data; a proposal does this for a research idea.

Learning Objectives

By the end of class, students can:

1. explain PCA as linear, usually lossy data compression;
2. center and standardize data before PCA when appropriate;
3. interpret principal directions, scores, loadings, and explained variance;
4. create and critique a 2D PCA visualization;
5. distinguish a motivating spark from a researchable problem;
6. identify stakeholders and make a specific request for support;
7. explain why proposals and papers receive peer review; and
8. produce a realistic six-week project plan for Weeks 17–22.

120-Minute Flow

1. **0–12 min:** the suitcase challenge—why compress data?
2. **12–35 min:** PCA geometry, centering, variance, and principal directions.
3. **35–52 min:** project, reconstruct, and measure information retained.
4. **52–58 min:** 2D visualization activity and interpretation warnings.
5. **58–63 min:** break.
6. **63–75 min:** launch the revised six-week research sprint.
7. **75–91 min:** find the spark, formulate the problem, and bound the scope.
8. **91–105 min:** stakeholders, proposals, and requests for support.
9. **105–116 min:** proposal and paper peer review; committee simulation.
10. **116–120 min:** proposal seed, next checkpoint, and homework briefing.

PCA: Compression by Choosing Better Coordinates

The Suitcase Challenge

A smart meter records 40 features every hour: voltage, current, power factor, harmonics, temperatures, weather, calendar signals, and more.

Challenge

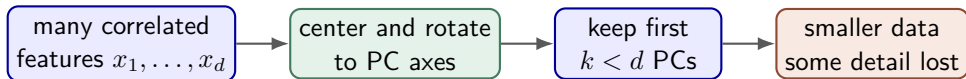
You may keep only two numbers per hour. Which two numbers would preserve the most visible structure in the data?

Keeping two original columns may throw away relationships distributed across many columns. PCA instead constructs two new coordinates from weighted combinations of all features.

PCA in One Sentence

Principal Component Analysis

PCA rotates centered data to new perpendicular axes, orders those axes by captured variance, and keeps only the first few axes when compression is desired.



Compression: Original Coordinates versus PCA Coordinates

Suppose one observation has $d = 100$ measurements. PCA can represent it approximately with $k = 2$ scores:

$$x \in \mathbb{R}^{100} \longrightarrow z = W_k^T (x - \mu) \in \mathbb{R}^2.$$

The compressed point can be expanded back into 100 dimensions:

$$\hat{x} = \mu + W_k z.$$

Lossy does not mean useless

If the first two directions capture most important variation, $\hat{x}$ can be close to x . The discarded directions are the reconstruction error.

Why PCA Is Useful for 2D and 3D Visualization

Humans cannot directly inspect a 40-dimensional cloud. PCA creates a 2D or 3D view that often preserves the largest patterns.

What the plot may reveal

- clusters or operating regimes;
- continuous gradients;
- unusual observations;
- overlap between labeled groups.

What the plot cannot prove

- a cluster is a real category;
- separation is causal;
- discarded dimensions are unimportant;
- a downstream model will succeed.

First Move: Center the Data

For n observations and d features, collect rows in $X \in \mathbb{R}^{n \times d}$. For feature j ,

$$\mu_j = \frac{1}{n} \sum_{i=1}^n X_{ij}, \quad X_{c,ij} = X_{ij} - \mu_j.$$

Why center?

PCA should describe variation around the center of the cloud, not the cloud's distance from the origin.

After centering, every feature has mean zero, up to numerical rounding.

Should We Standardize Too?

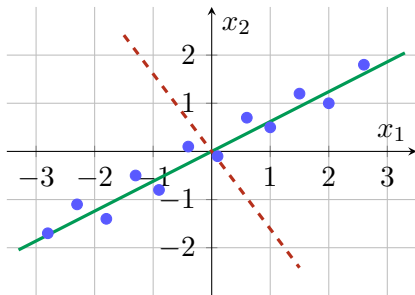
Standardization divides each centered feature by its standard deviation:

$$X_{s,ij} = \frac{X_{ij} - \mu_j}{s_j}.$$

Situation	Reasonable choice
Features use different units	Usually standardize, so volts do not dominate percentages by scale alone.
Same units and magnitude matters	Centering without scaling may preserve meaningful magnitude differences.
Near-constant feature	Remove or investigate it; dividing by a tiny standard deviation is unstable.

The preprocessing choice changes the question PCA answers.

PCA Searches for the Direction of Greatest Spread



PC1

The green direction captures the largest projected variance.

PC2

The red direction is perpendicular to PC1 and captures the largest remaining variance.

A Projection Turns a Point into a Score

Let v_1 be a unit vector pointing along PC1. The PC1 score of centered observation x_c is

$$z_1 = x_c^T v_1.$$

With the first k principal directions as columns of W_k ,

$$Z = X_c W_k.$$

- Each row of Z is one compressed observation.
- Each column of Z contains the scores for one principal component.
- The entries of W_k are often called **loadings**; they describe how original features contribute to each component.

Where Do the Principal Directions Come From?

Estimate the covariance matrix of centered data:

$$C = \frac{1}{n-1} X_c^T X_c.$$

Then solve

$$Cv_j = \lambda_j v_j, \quad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0.$$

Interpretation

The eigenvector v_j is a principal direction. Its eigenvalue λ_j is the variance captured along that direction.

In reliable software, PCA is commonly computed using singular value decomposition rather than explicitly forming the covariance matrix.

Explained Variance Measures the Compression Tradeoff

The explained-variance ratio of component j is

$$r_j = \frac{\lambda_j}{\sum_{m=1}^d \lambda_m}.$$

Keeping the first k components retains the fraction

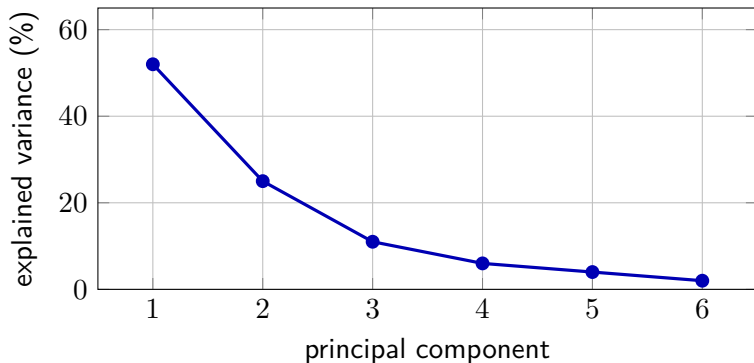
$$R_k = \sum_{j=1}^k r_j.$$

Example

If the first two components explain 58% and 24%, the 2D view retains 82% of total variance and discards 18%.

A target such as 90% or 95% is a decision rule, not a universal law.

The Scree Plot Helps Choose the Number of Components



Look for diminishing returns, then check whether the retained representation is adequate for the actual task.

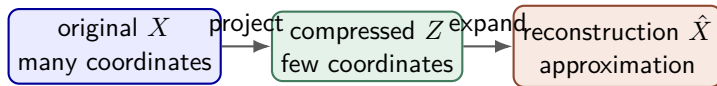
Reconstruction Makes Information Loss Visible

Compress and reconstruct:

$$Z = X_c W_k, \quad \hat{X} = Z W_k^T + \mu.$$

A simple reconstruction measure is

$$\text{MSE}_{\text{recon}} = \frac{1}{nd} \sum_{i=1}^n \sum_{j=1}^d (X_{ij} - \hat{X}_{ij})^2.$$



Manual Mini-Challenge: Find the Long Direction

Consider four already centered observations:

$$X_c = \begin{bmatrix} -2 & -1 \\ -1 & -1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}.$$

In pairs:

1. sketch the four points;
2. predict whether PC1 is closer to $(1, 0)$, $(0, 1)$, or $(1, 1)$;
3. explain the prediction using spread, not labels;
4. decide whether one component would reconstruct the cloud well.

PCA in a Small, Reproducible Pipeline

```
1 from sklearn.preprocessing import StandardScaler
2 from sklearn.decomposition import PCA
3
4 X_scaled = StandardScaler().fit_transform(X)
5
6 pca = PCA(n_components=2)
7 Z = pca.fit_transform(X_scaled)
8
9 print(pca.explained_variance_ratio_)
10 print(pca.components_)           # rows are principal directions
11
12 X_scaled_hat = pca.inverse_transform(Z)
```

Fit preprocessing and PCA on training data only if the representation will be used inside a predictive evaluation pipeline.

How to Read a PCA Scatter Plot

For every visual claim, report:

1. what one point represents;
2. what preprocessing was applied;
3. the explained variance on each displayed axis;
4. what color, shape, or size encodes;
5. one visible pattern; and
6. one alternative explanation or limitation.

Do not name a component too quickly

A component is a weighted combination. Inspect its largest positive and negative loadings before giving it a human-friendly interpretation.

Common PCA Failure Modes

- **Scale dominance:** a large numerical unit controls the components.
- **Outlier rotation:** a few extreme points pull the principal directions.
- **Linear blindness:** a curved low-dimensional structure may not flatten well with linear axes.
- **Variance is not value:** a low-variance feature may still predict a rare critical event.
- **Unsupervised objective:** PCA ignores labels when choosing directions.
- **Visual overclaiming:** apparent clusters can change with preprocessing or omitted dimensions.
- **Leakage:** fitting PCA before a train/test split lets test information influence the representation.

PCA Checkpoint

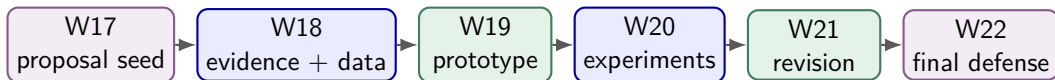
Decide whether each statement is true or false, then repair the false statements.

1. PC1 is the original feature with the largest variance.
2. The first two PCs are perpendicular.
3. A 2D PCA plot always preserves all important information.
4. Standardization is a modeling decision, not a ceremonial step.
5. PCA uses class labels to maximize separation.
6. Explained variance and reconstruction error describe related views of information retention.

The Six-Week Demo Research Project

Program Update: Six Weeks Remain

The program is now **22 weeks**, not 20. Week 17 begins a six-week research sprint:



Target

A small but complete demo project with the structure and care expected of a strong bachelor's graduation project: question, evidence, method, results, limitations, reproducible artifact, and presentation.

What Counts as a Complete Small Research Project?

1. A clear problem and motivation.
2. A bounded research question or claim.
3. Relevant prior work and a credible gap.
4. Data, materials, or evidence that can actually be accessed.
5. A method another person could understand and repeat.
6. A baseline or comparison.
7. Results, including negative or mixed results.
8. Limitations and responsible interpretation.
9. A final artifact: report, code, data notes, and presentation.

Completeness matters more than pretending the project is enormous.

From Excitement to a Researchable Problem

Research Often Starts with a Spark

The starting point may be an observation, frustration, curiosity, contradiction, or sudden impulse:
"I need to understand this."

First instruction

Find something that genuinely excites you. Do not demand a perfect method, dataset, and argument from the first minute.

Second instruction

The spark gives energy, not automatic validity. The idea must still survive evidence, feasibility checks, ethical reflection, and criticism.

Three Lists Find Better Project Seeds

Write quickly without judging each item:

Interests

What topics, communities, systems, or tools make you curious?

Resources

What data, software, equipment, mentors, and prior skills can you access?

Constraints

What can you ethically and realistically finish in six weeks?

Strong candidates live near the overlap, but excitement is allowed to lead the search.

Topic, Problem, Question, and Deliverable Are Different

Element	Example
Topic	Residential electricity demand.
Problem	Short peak events are hard to detect early from ordinary meter summaries.
Question	Can features from the previous 30 minutes improve detection of the next-hour peak compared with a historical-average baseline?
Deliverable	A reproducible notebook, comparison table, error analysis, and short report.

A topic names an area. A research problem identifies a consequential uncertainty or failure inside that area.

A Useful Problem Statement Has Five Parts

Complete this structure:

1. **Context:** In what system or situation does this occur?
2. **Observed problem:** What is unknown, unreliable, inefficient, unfair, or unexplained?
3. **Affected stakeholder:** Who experiences the consequence?
4. **Consequence:** Why does the uncertainty matter?
5. **Evidence gap:** What do we still need to measure, compare, build, or understand?

Sentence frame

In [context], [stakeholder] faces [problem], which matters because [consequence]. Existing knowledge or tools do not yet establish [gap].

Turn the Problem into a Research Question

A six-week question should name:

object + relationship or comparison + evidence + boundary.

Too broad

How can AI improve the smart grid?

Testable and bounded

On one public hourly-load dataset, does a random-forest model using weather and calendar features reduce peak-hour MAE relative to a seasonal-average baseline?

A good question can be answered even if the preferred method fails.

Scope Is a Scientific Skill

Shrink a project by limiting one or more dimensions:

- one population, site, dataset, or simulation;
- one main outcome;
- one or two methods plus a baseline;
- one time window;
- one measurable claim;
- one reproducible demonstration instead of a deployable product.

Bachelor-level does not mean industry-scale

The project earns strength through a complete argument, honest evaluation, and clear limitations—not through the largest dataset or most fashionable model.

The Minimum Viable Research Project

Define three layers before building:

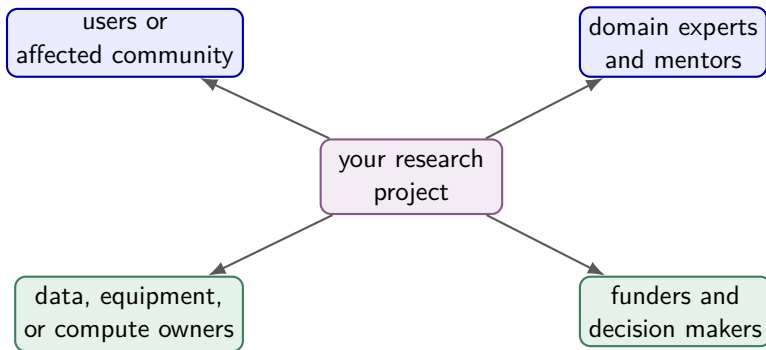
Layer	Meaning
Must have	The smallest evidence needed to answer the question honestly.
Should have	One improvement that makes the comparison or explanation stronger.
Could have	An extension attempted only after the must-have result exists.

If the must-have layer cannot be completed in four working weeks, the scope is still too large; Week 21 is needed for revision and Week 22 for presentation.

Stakeholders and the Purpose of a Proposal

Who Are the Stakeholders?

A stakeholder can affect the project or be affected by it.



Different stakeholders judge value, risk, evidence, and clarity differently.

A Proposal Is an Argument for Support

A proposal helps another person decide whether to provide:

- money or compute;
- data, equipment, or access;
- expert advice or supervision;
- participant time or community trust;
- permission, partnership, or technical support.

The proposal's central promise

This problem matters; this question is answerable; this plan is credible; the requested support has a clear purpose; and the expected evidence will be worth the cost and risk.

The First One-Page Proposal

Version 0.1 should contain:

1. working title and one-sentence spark;
2. problem statement and main research question;
3. why the answer matters and to whom;
4. two to four relevant pieces of prior evidence to find;
5. data or materials and access status;
6. proposed method, baseline, and evaluation measure;
7. six-week milestones and smallest complete result;
8. main risks, ethics, and limitations;
9. named stakeholders; and
10. one specific request for help.

This is a discussion object, not a permanent contract.

Make the Request Specific

Weak request

“Please support my AI project.”

Reviewable request

“I request two 30-minute meetings with a power-systems mentor to check whether my peak-event definition is physically meaningful, plus access to the public dataset documentation.”

A strong request names the resource, quantity, timing, purpose, and decision it will improve.

Why Peer Review Exists

The Evaluation Problem

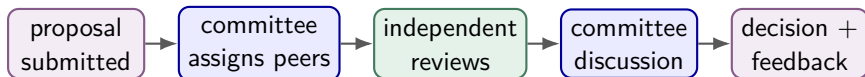
Imagine controlling limited money or resources. A specialist proposes work in a field you do not know deeply. The specialist may have spent years on the topic.

The hard question

How can a decision maker judge whether the idea is important, technically credible, feasible, and worth supporting without pretending to personally master every field?

Peer review delegates detailed evaluation to multiple people with relevant expertise and asks them to make their reasoning visible.

A Simplified Proposal Peer-Review Process



Reviewers do not need to predict the future perfectly. They compare claims with evidence, expose risks, and make the decision more accountable.

Proposal Review and Paper Review Ask Different Questions

	Proposal review	Paper review
Timing	Before most work is done	After results are produced
Main decision	Should resources or permission be given?	Is the reported work credible and suitable to publish?
Evidence available	Plan, prior work, pilot evidence, team, budget	Methods, data, analysis, results, limitations
Central risks	Importance, feasibility, ethics, execution	Validity, novelty, reproducibility, interpretation
Useful outcome	Better plan, support decision, or rejection	Revision, acceptance, or rejection

A Committee Needs an Explicit Rubric

Score each category from 1 (weak) to 5 (excellent):

1. **Significance:** does the problem matter to named stakeholders?
2. **Question clarity:** is the claim bounded and answerable?
3. **Evidence grounding:** does prior work support the gap?
4. **Method fit:** can the proposed evidence answer the question?
5. **Feasibility:** can this student finish with available time and resources?
6. **Evaluation:** is there a baseline and a success measure?
7. **Ethics and risk:** are harms, privacy, and failure modes considered?
8. **Communication:** is the request understandable and specific?

Peer Review Is Useful, Not Magical

Committees can still suffer from:

- reviewer bias, conflicts of interest, and status effects;
- preference for familiar ideas over risky new ones;
- inconsistent standards or missing expertise;
- excessive confidence in polished writing;
- limited reviewer time; and
- disagreement about value or acceptable risk.

Better practice

Use multiple reviewers, disclose conflicts, score explicit criteria, separate major concerns from preferences, invite revision, and preserve room for reasoned disagreement.

How to Give a Useful Review

Use four moves:

1. **Restate:** “I understand the project to ask...”
2. **Strength:** name one specific reason the project could matter or work.
3. **Major concern:** identify the highest-impact threat to validity or completion.
4. **Actionable revision:** propose a test, clarification, scope cut, baseline, or stakeholder conversation.

Review the work, not the person

Replace “You do not understand the topic” with “The proposal does not yet explain why this measurement represents the claimed outcome.”

Committee Simulation

Each student gives a 60-second pitch containing:

1. the spark;
2. the problem and affected stakeholder;
3. the research question;
4. the smallest complete result; and
5. the request for help.

Reviewers respond with:

- one strength;
- one major concern;
- one question whose answer could change the decision; and
- one recommended next action.

Plan, Iterate, and Finish

The Week 17–22 Research Rhythm

Week	Main question	Required evidence	Checkpoint
17	What should I study?	idea bank, problem, stakeholders, feasibility	proposal v0.1 + committee review
18	What is already known?	focused literature map, data access, refined gap	proposal v0.2 + data card
19	Can the method run end to end?	baseline and minimum prototype	reproducible demo v0.1
20	What do the comparisons show?	planned experiments, metrics, error analysis	results table + figures
21	What survives criticism?	peer review, robustness check, limitations	report and demo revision
22	What did I learn and contribute?	final artifact, claims, evidence, reflection	presentation and defense

The mentor and student revise the idea, plan, and stakeholder map together at every checkpoint.

Risk Register: Plan for Failure Before It Arrives

For each risk, record probability, impact, warning sign, and response.

Risk	Early warning	Response
Data inaccessible	no usable sample by Week 18	switch to a named public fallback dataset
Method too complex	no baseline by Week 19	reduce to one method and one comparison
Metric mismatch	result does not answer stakeholder need	revise outcome definition with mentor
Weak signal	performance near baseline	report honestly and analyze failure modes
Schedule slip	two missed checkpoints	cut could-have features immediately

Today's Proposal Seed

Before leaving, write one sentence for each box:

1. I am excited by _____ because _____.
2. The observable problem is _____.
3. It affects _____ by _____.
4. My current research question is _____.
5. The smallest complete result is _____.
6. I need help from _____ with _____.

Circle the sentence with the greatest uncertainty. That becomes the first mentor conversation.

Exit Ticket

Answer briefly:

1. Why is PCA a compression method?
2. What does the first eigenvalue describe?
3. Why might standardization change a PCA plot?
4. Name one claim a 2D PCA plot cannot prove.
5. How is a research topic different from a research problem?
6. Name one stakeholder and one specific resource they could provide.
7. What is one difference between proposal review and paper review?
8. What must be ready at the end of Week 19?

Homework for This Week

PCA evidence

Implement the core PCA steps, verify them against scikit-learn, create a careful 2D visualization, and measure reconstruction loss.

Proposal v0.1

Build an idea bank, choose one exciting but feasible seed, formulate the problem, map stakeholders, and write a one-page proposal.

Peer-review checkpoint

Exchange proposals using the committee rubric, revise the plan, and submit a response-to-reviewers table plus the Week 17–22 milestone plan.

Key Takeaways

1. PCA replaces many correlated features with ordered, perpendicular components.
2. Keeping only a few components is usually lossy compression; explained variance and reconstruction error reveal the tradeoff.
3. PCA is a powerful 2D/3D inspection tool, but the projection can hide structure and does not prove causal or categorical claims.
4. Research can begin with excitement before every detail is reasonable.
5. A proposal transforms that spark into a bounded question, evidence plan, stakeholder argument, and specific request.
6. Peer review helps non-specialist decision makers use distributed expertise, while explicit rubrics and revision reduce some of its limitations.
7. Weeks 17–22 form one complete cycle: propose, ground, prototype, test, revise, and defend.